

```
{ "cells": [ { "cell_type": "markdown", "id": "e16ac040", "metadata": {}, "source": [ "# Scikit-learn Complete Guide – Part 1 of 2\n", "***Topics in this part:** 1) Basics 2) Preprocessing 3) Regression 4) Classification 5) Clustering 6) Dimensionality Reduction (PCA)\n", "\n", "***Part 2 (next):** Model Evaluation  Tuning  Pipelines  Ensembles  Saving Models\n", "\n", "***How to use:** run cells top to bottom. Read the comments, then change values (test size, `k`, `alpha`...) and see what happens. All datasets are built into scikit-learn, so no downloads are needed." ] }, { "cell_type": "code", "execution_count": 1, "id": "82c143b7", "metadata": { "execution": { "iopub.execute_input": "2026-09-28T11:59:44.699991Z", "iopub.status.busy": "2026-09-28T11:59:44.699790Z", "iopub.status.idle": "2026-09-28T11:59:45.900543Z", "shell.execute_reply": "2026-09-28T11:59:45.899228Z" } }, "outputs": [], "source": [ "import numpy as np\n", "import pandas as pd\n", "import matplotlib.pyplot as plt\n", "import warnings\n", "warnings.filterwarnings('ignore')\n", "\n", "np.set_printoptions(precision=3, suppress=True)\n", "pd.set_option('display.precision', 3)" ] }, { "cell_type": "markdown", "id": "6aa66035", "metadata": {}, "source": [ "---\n", "# 1. Basics: estimator API, datasets, train_test_split\n", "\n", "Every scikit-learn object follows the same few methods:\n", "\n", "| Method | Used by | Meaning |\n", "|---|---|---|\n", "| `fit(X, y)` | all models / transformers | learn from data |\n", "| `predict(X)` | models | make predictions |\n", "| `transform(X)` | transformers (scalers, PCA...) | change the data using what was learned in `fit` |\n", "| `fit_transform(X)` | transformers | `fit` + `transform` in one step |\n", "| `score(X, y)` | models | quick default metric (accuracy for classifiers,  $R^2$  for regressors) |\n", "\n", "***Convention:** `X` = features (2D: rows — columns), `y` = target (1D)." ] }, { "cell_type": "code", "execution_count": 2, "id": "5a928c38", "metadata": { "execution": { "iopub.execute_input": "2026-09-28T11:59:45.902697Z", "iopub.status.busy": "2026-09-28T11:59:45.902108Z", "iopub.status.idle": "2026-09-28T11:59:46.782210Z", "shell.execute_reply": "2026-09-28T11:59:46.781295Z" } }, "outputs": [ { "name": "stdout", "output_type": "stream", "text": [ "X shape: (150, 4)\n", "y shape: (150,)\n", "Features: ['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal width (cm)']\n", "Classes : [np.str_('setosa'), np.str_('versicolor'), np.str_('virginica')]\n" ] }, { "data": { "text/html": [ "\n", "
```